



4. Program 2: Hotel Booking System

4.1 Introduction and Background

A Hotel Booking System simplifies room reservation by managing room availability, bookings, and check-outs. This project develops a console-based Python application using in-memory data structures with a consistent boxed terminal interface.

4.2 Objectives

- Develop a menu-driven hotel booking system.
- Manage multiple room categories and availability.
- Generate unique booking IDs.
- Calculate booking cost based on room type and stay duration.
- Support booking, check-out, and booking history.
- Build a consistent terminal interface.

4.3 Problem Statement / Driving Question

How can a hotel manage room availability, bookings, check-outs, and billing using a simple console-based application without an external database?

4.4 Methodology

- **Program Design:** Created a boxed terminal interface using reusable helper functions and an ASCII-art banner.
- **Data Structures:** Used ROOM_TYPES to store room details and BOOKINGS to manage reservation records.
- **Core Booking Functions:** Implemented room availability checking, booking creation with unique booking IDs, total cost calculation, and check-out functionality.
- **Menu-Driven Control Flow:** Built a continuous menu with room viewing, booking, check-out, and booking history options, including input validation and error handling.

4.5 Expected Outcomes / Deliverables

- A functional hotel booking system.
- Real-time room availability tracking.
- Unique booking ID generation.
- Accurate booking cost calculation.
- Input validation and error handling.
- A user-friendly boxed terminal interface.